

Ex.no:8	MINI PROJECT – HEALTHCARE APPOINTMENT ANALYSIS

Domain and Problem Statement:

Domain: Healthcare Analytics & Appointment Management

Problem Statement: Healthcare organizations face challenges due to missed medical appointments, which can reduce healthcare service efficiency, waste available appointment slots, and affect patient management. This project addresses these challenges by analyzing medical appointment records to identify factors associated with appointment no-shows. The project transforms raw appointment data into meaningful business intelligence using Python, SQL Server, Power BI and Tableau to evaluate attendance patterns based on age, gender, SMS reminders, waiting time, scholarship status, health conditions and neighbourhood.

Dataset Description:

Dataset Name: Medical Appointment No-Show Dataset.

Granularity: Each row represents a single medical appointment record.

Timeframe Covered: May 2016.

Original Records: 110,527 appointment records.

Cleaned Records: 110,521 appointment records.

Key Entities and Attributes:

Patient Information:

- PatientId
- Gender
- Age
- Neighbourhood

Appointment Information:

- AppointmentID
- ScheduledDay
- AppointmentDay
- Waiting_Days
- Waiting_Time_Group

Health Information:

- Hipertension
- Diabetes
- Alcoholism
- Handcap
- Health_Conditions

Appointment Support Information:

- Scholarship
- SMS_received
- SMS_Status

Appointment Outcome:

- No_show
- Attendance_Status

Derived Attributes:

- Age_Group
- Waiting_Days
- Waiting_Time_Group
- Health_Conditions
- SMS_Status
- Attendance_Status

Data Source:

Primary Source: Kaggle Open Dataset – Medical Appointment No-Show Dataset.

Original File: KaggleV2-May-2016.csv

Data Processing: Python using Pandas.

Storage Infrastructure: Cleaned appointment records stored in a local Microsoft SQL Server database.

Database Name: HealthcareAnalytics

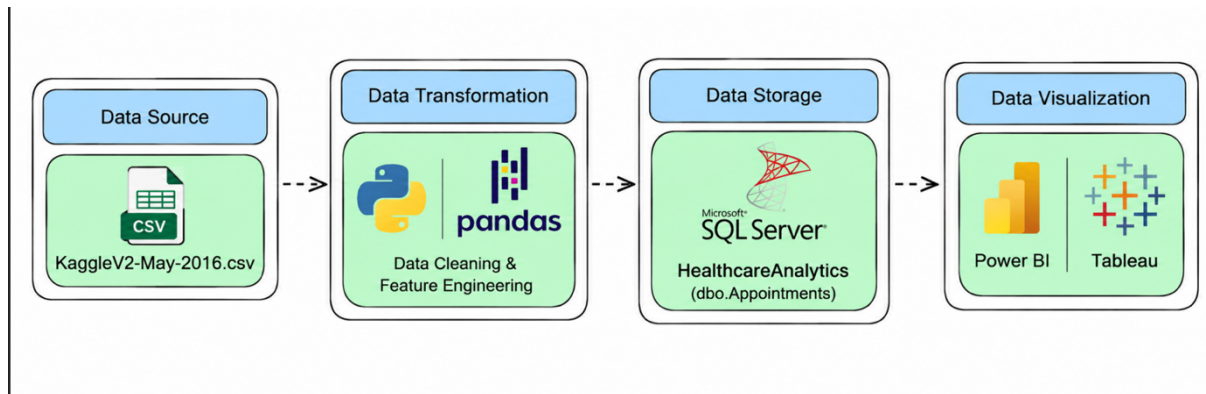
Table Name: dbo.Appointments

Visualization Tools:

- Microsoft Power BI
- Tableau Desktop

The project therefore follows the same source → transformation → database → visualization approach used in the reference document, but with the healthcare appointment dataset and SQL Server architecture.

Architecture:



Program:

1. Python – Data Cleaning & Transformation

```
import pandas as pd
```

```
df = pd.read_csv("KaggleV2-May-2016.csv")
```

```
# Create a copy for processing
```

```
healthcare = df.copy()
```

```
# Handle missing values
```

```
healthcare["Gender"] = healthcare["Gender"].fillna("Unknown")
```

```
healthcare["Neighbourhood"] = healthcare["Neighbourhood"].fillna("Unknown")
```

```
healthcare["No-show"] = healthcare["No-show"].fillna("Unknown")
```

```
# Replace missing numerical values with 0
```

```
numeric_columns = [  
    "Age",  
    "Scholarship",  
    "Hipertension",  
    "Diabetes",  
    "Alcoholism",  
    "Handcap",  
    "SMS_received"  
]  
  
for column in numeric_columns:  
    healthcare[column] = healthcare[column].fillna(0)  
  
# Remove duplicate records  
healthcare.drop_duplicates(inplace=True)  
  
# Convert date columns  
healthcare["ScheduledDay"] = pd.to_datetime(  
    healthcare["ScheduledDay"], errors="coerce"  
)  
  
healthcare["AppointmentDay"] = pd.to_datetime(  
    healthcare["AppointmentDay"], errors="coerce"  
)  
  
# Create Waiting Days  
healthcare["Waiting_Days"] = (  
    healthcare["AppointmentDay"].dt.normalize()  
    - healthcare["ScheduledDay"].dt.normalize()  
)dt.days  
  
# Create Attendance Status
```

```
healthcare["Attendance_Status"] = healthcare["No-show"].map({  
    "No": "Attended",  
    "Yes": "No-Show"  
})
```

Create SMS Status

```
healthcare["SMS_Status"] = healthcare["SMS_received"].map({  
    0: "No SMS",  
    1: "SMS Received"  
})
```

Create Health Conditions

```
healthcare["Health_Conditions"] = (  
    healthcare["Hipertension"]  
    + healthcare["Diabetes"]  
    + healthcare["Alcoholism"]  
)
```

Create Age Group

```
def age_group(age):  
    if age <= 17:  
        return "0-17"  
    elif age <= 30:  
        return "18-30"  
    elif age <= 45:  
        return "31-45"  
    elif age <= 60:  
        return "46-60"  
    elif age <= 75:  
        return "61-75"  
    else:  
        return "76+"
```

```
healthcare["Age_Group"] = healthcare["Age"].apply(age_group)
```

```
# Create Waiting Time Group
```

```
def waiting_time_group(days):
```

```
    if days == 0:
```

```
        return "Same Day"
```

```
    elif days <= 7:
```

```
        return "1-7 Days"
```

```
    elif days <= 14:
```

```
        return "8-14 Days"
```

```
    elif days <= 30:
```

```
        return "15-30 Days"
```

```
    elif days <= 60:
```

```
        return "31-60 Days"
```

```
    else:
```

```
        return "60+ Days"
```

```
healthcare["Waiting_Time_Group"] = healthcare["Waiting_Days"].apply(
```

```
    waiting_time_group
```

```
)
```

```
# Standardize column names
```

```
healthcare.columns = (
```

```
    healthcare.columns
```

```
    .str.lower()
```

```
    .str.strip()
```

```
    .str.replace(" ", "_")
```

```
)
```

```
# Display dataset information
```

```
print("Original Shape:", df.shape)
```

```
print("Cleaned Shape:", healthcare.shape)
```

```
print("\nDataset Preview:")
```

```
print(healthcare.head())
```

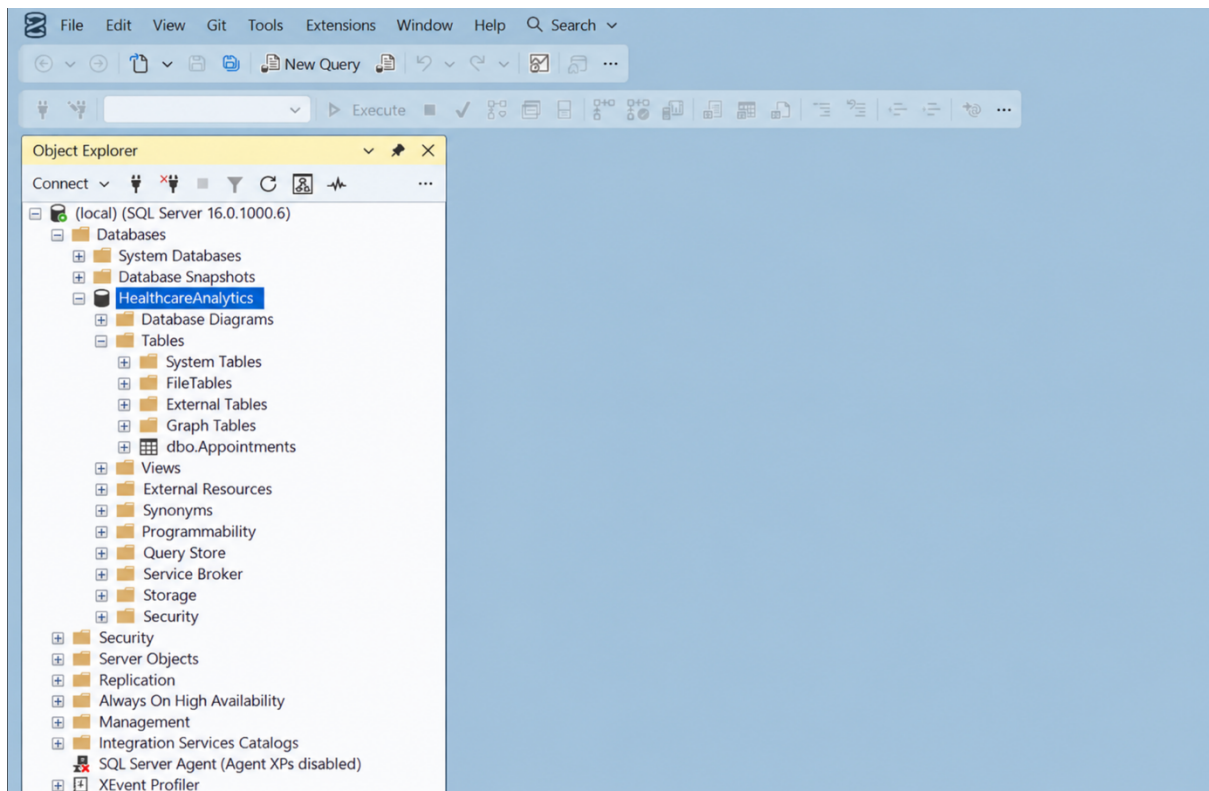
```
print("\nDataset Information:")
```

```
print(healthcare.info())
```

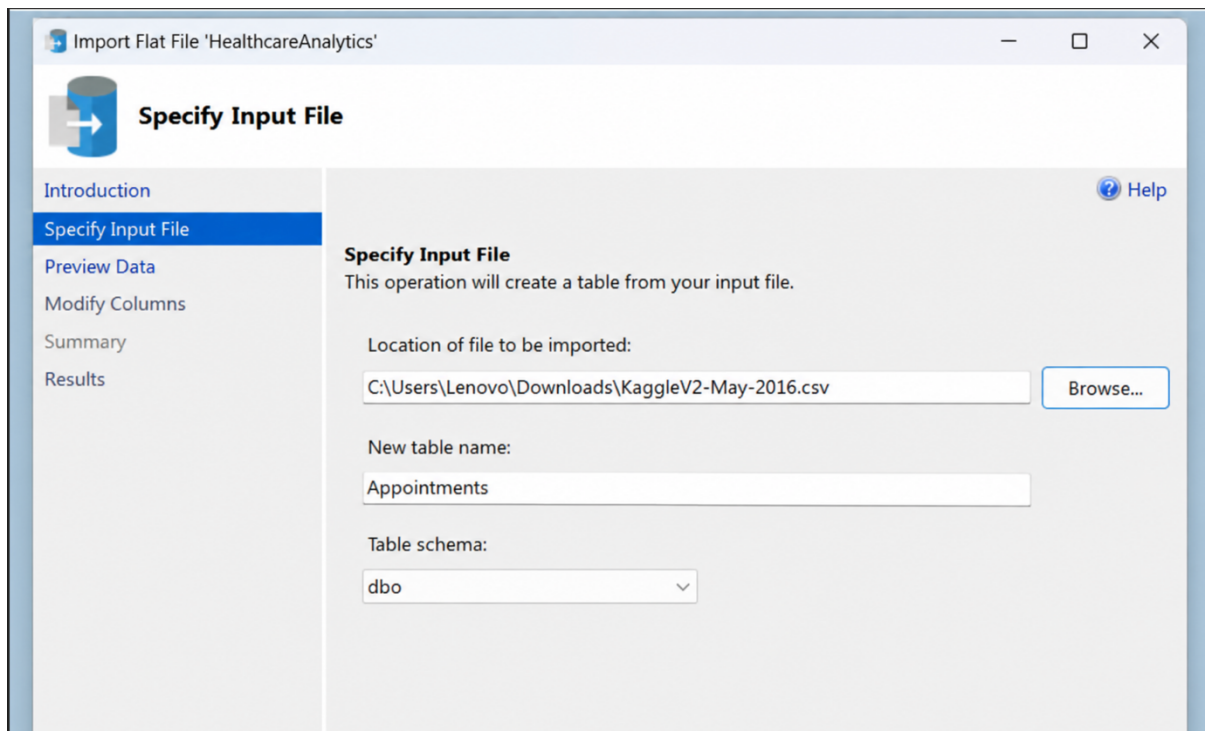
2. SQL SERVER CONNECTIVITY

INSTALL SQL SERVER MANAGEMENT STUDIO

1) Create database named “dashboardDB”



2) Import the csv file of the dataset

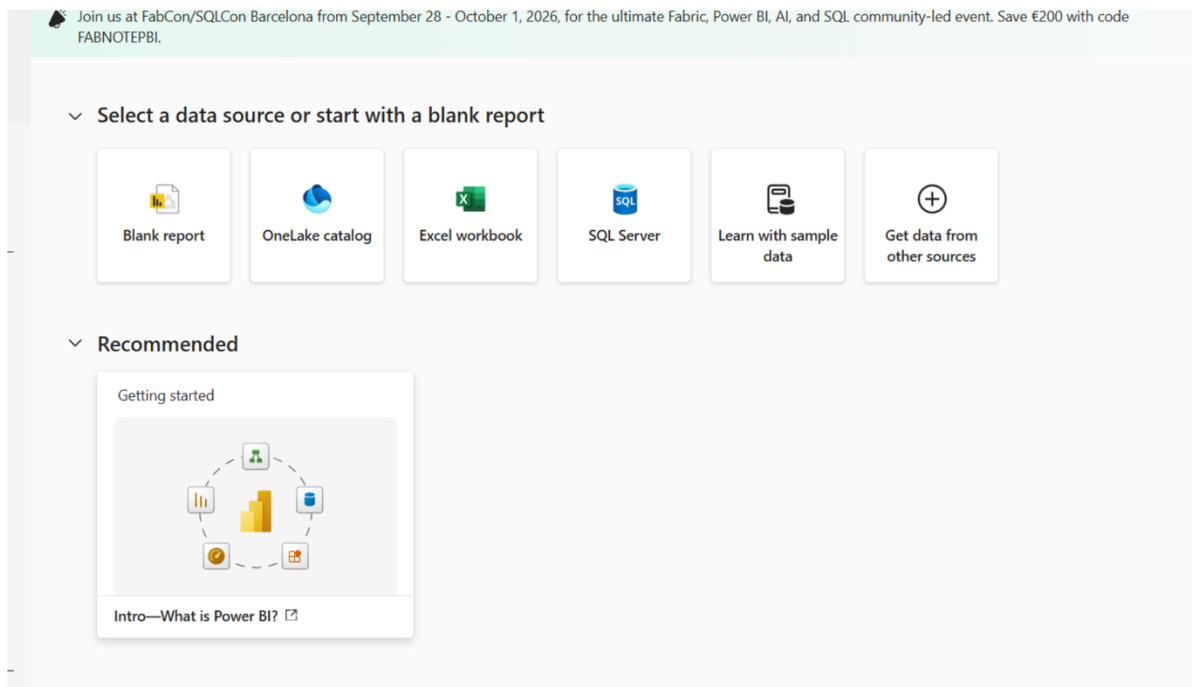


The screenshot shows the 'Import Flat File' wizard in Power BI. The window title is 'Import Flat File 'HealthcareAnalytics''. The left sidebar has a menu with 'Introduction', 'Specify Input File' (selected), 'Preview Data', 'Modify Columns', 'Summary', and 'Results'. The main area is titled 'Specify Input File' and contains the following fields:

- Location of file to be imported:** A text box containing 'C:\Users\Lenovo\Downloads\KaggleV2-May-2016.csv' and a 'Browse...' button.
- New table name:** A text box containing 'Appointments'.
- Table schema:** A dropdown menu with 'dbo' selected.

POWER BI

3) Now go to MICROSOFT POWER BI



4) Click SQL SERVER

SQL Server database

Server ⓘ
localhost

Database (optional)
HealthcareAnalytics

Data Connectivity mode ⓘ
☒ Import
☐ DirectQuery

▸ Advanced options

OKCancel

5) Click OK and click transform data

TXN000000023	CUST018432	01-01-2019 00:00:00	21:28	Cl
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LoadTransform DataCancel

6) Create DAX measures

TOTAL APPOINTMENTS:

```
1 Total Appointments = COUNT(Appointments[AppointmentID])
```

TOTAL ATTENDED :

```
1 Total Attended =  
2 CALCULATE(  
3     [Total Appointments],  
4     Appointments[No_show] = "No"  
5 )
```

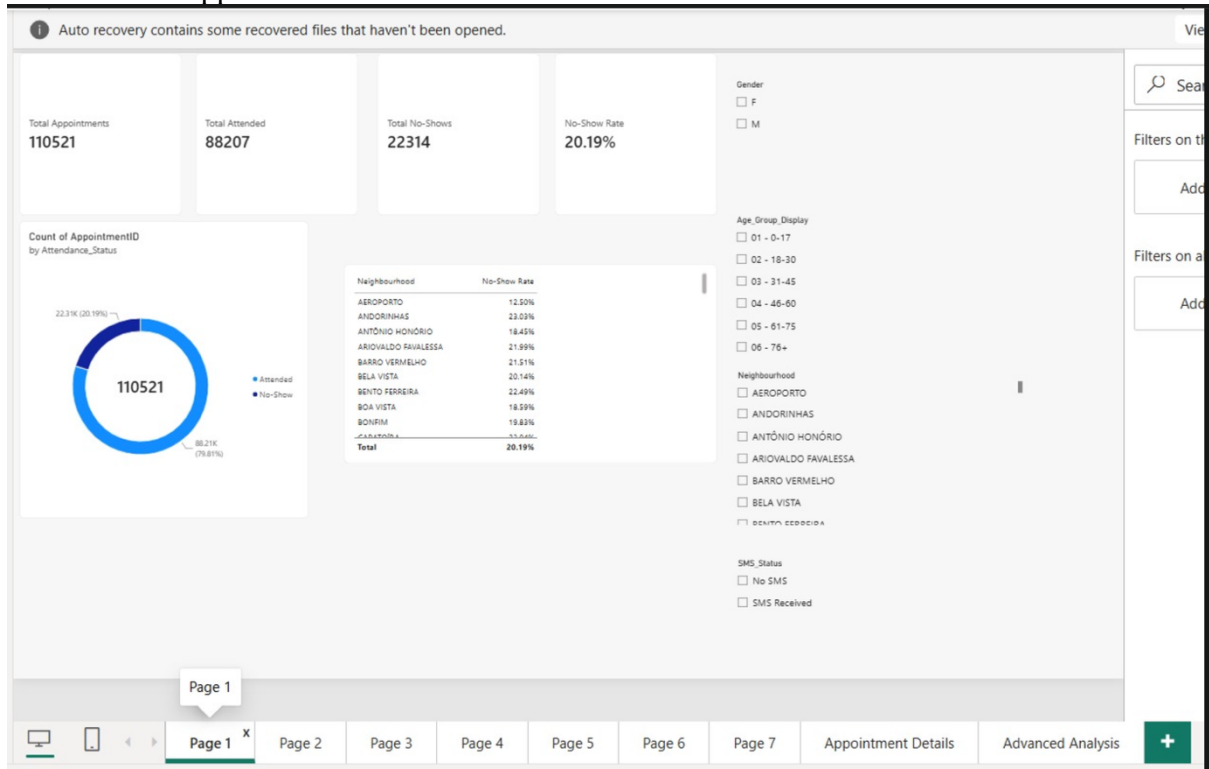
TOTAL NO-SHOWS:

```
1 Total No-Shows =  
2 CALCULATE(  
3     [Total Appointments],  
4     Appointments[No_show] = "Yes"  
5 )
```

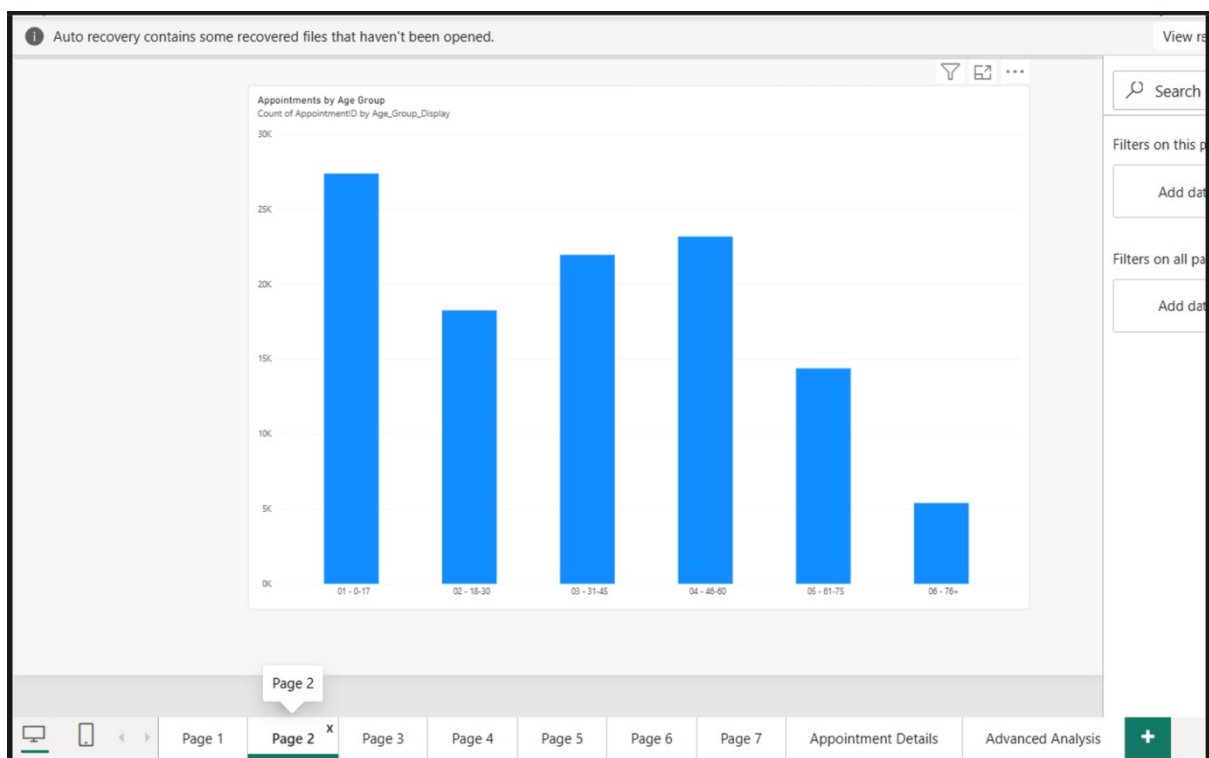
NO-SHOW RATE:

```
1 No-Show Rate =  
2 DIVIDE(  
3     [Total No-Shows],  
4     [Total Appointments],  
5     0  
6 )
```

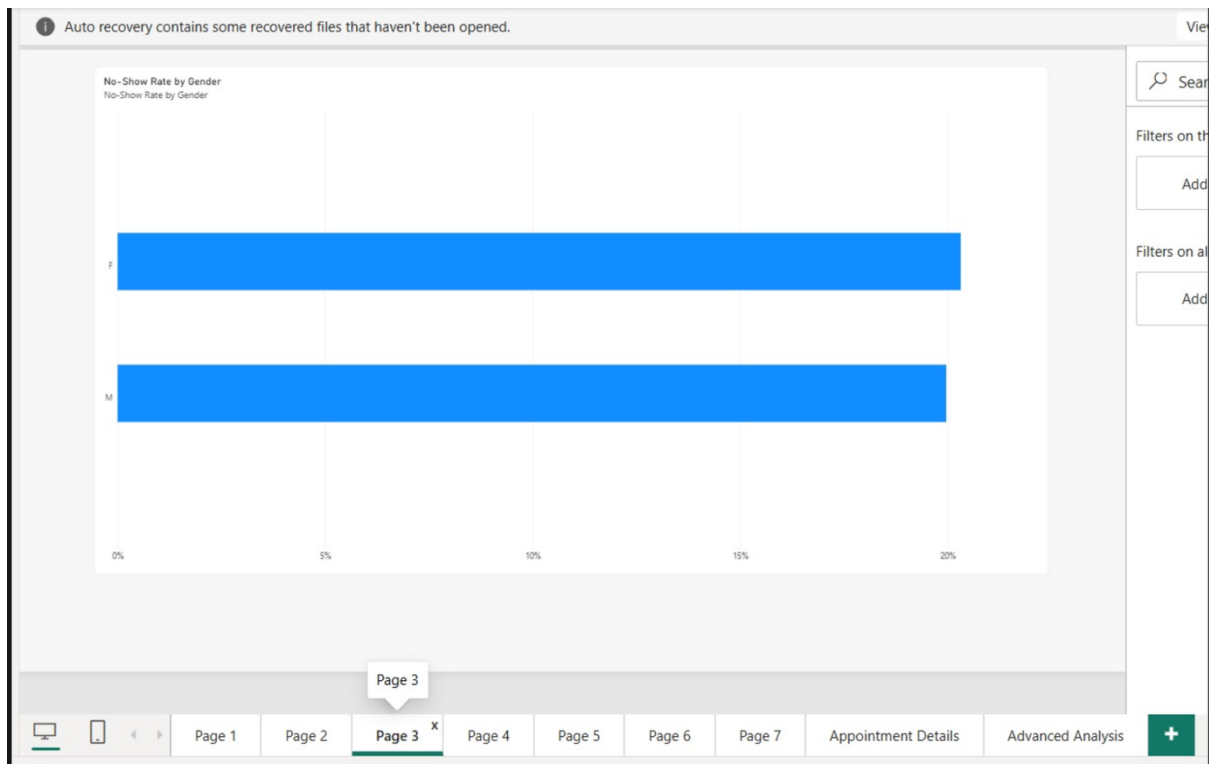
1. Healthcare Appointment Overview



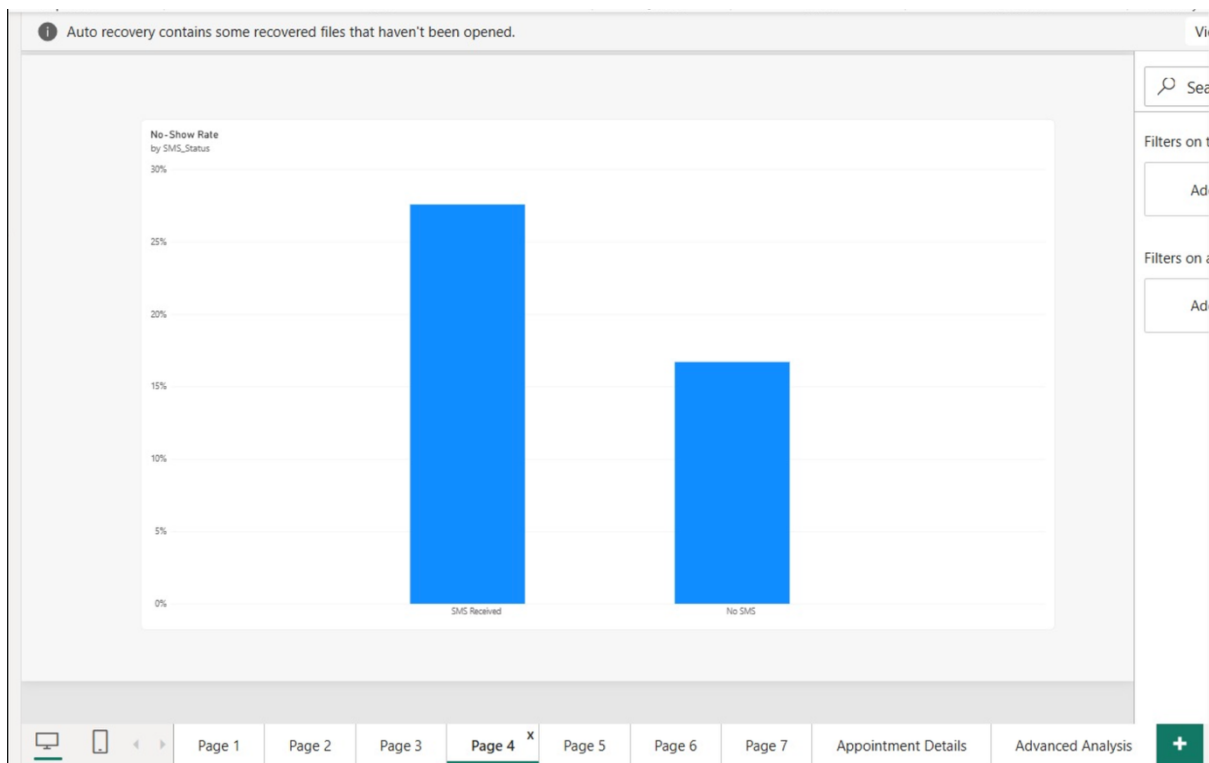
2. Appointment Analysis by Age Group



3. No-Show Rate by Gender



4. No-Show Rate by SMS Status



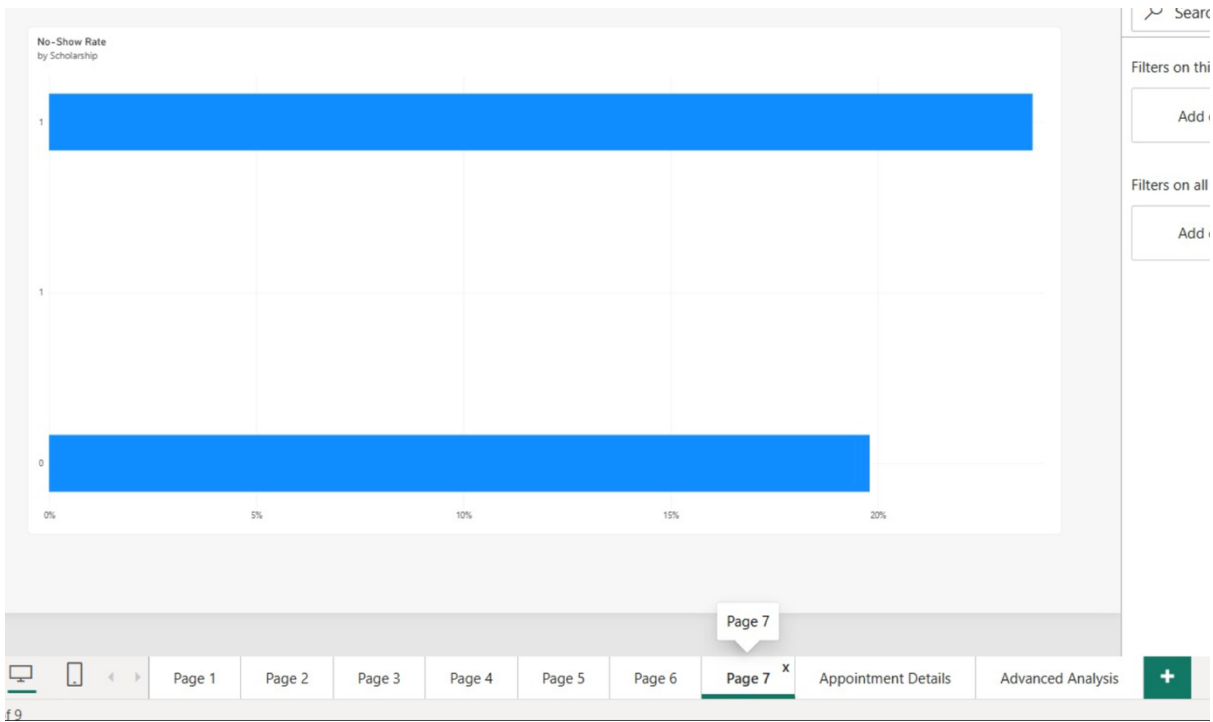
5. No-Show Rate by Waiting Time



6. No-Show Rate by Health Conditions



7. No-Show Rate by Scholarship



8. Appointment Details – Drill-Through Analysis

Auto recovery contains some recovered files that haven't been opened.

View

Search

Filters on this page

Add

Filters on all pages

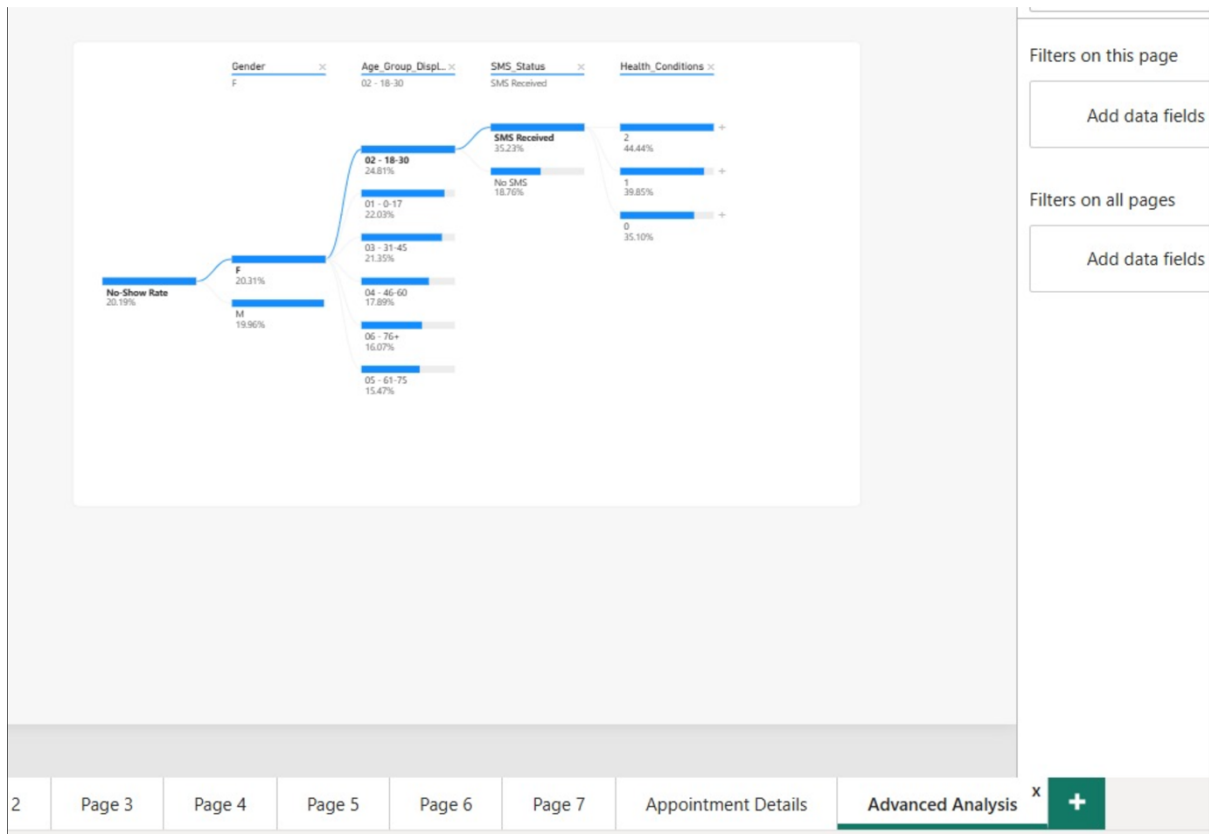
Add

AppointmentID	Age	Neighbourhood	Waiting_Days	SMS_Status	Attendance_Status	Gender	AppointmentDay
S284407	82	BONFRM	105	No SMS	Attended	M	5/9/2016 9:30:00 AM
S361416	75	BONFRM	72	SMS Received	No-Show	M	4/29/2016 9:30:00 AM
S376913	35	BONFRM	71	SMS Received	Attended	F	5/3/2016 9:30:00 AM
S399350	28	BONFRM	83	No SMS	Attended	M	5/19/2016 9:30:00 AM
S408322	39	BONFRM	81	No SMS	Attended	M	5/20/2016 9:30:00 AM
S412148	67	BONFRM	78	No SMS	No-Show	M	5/18/2016 9:30:00 AM
S412152	67	BONFRM	78	No SMS	No-Show	M	5/18/2016 9:30:00 AM
S511476	8	BONFRM	32	SMS Received	Attended	M	4/29/2016 9:30:00 AM
S511566	48	BONFRM	32	SMS Received	Attended	F	4/29/2016 9:30:00 AM
S511720	78	BONFRM	32	No SMS	Attended	M	4/29/2016 9:30:00 AM
S514447	72	BONFRM	67	SMS Received	Attended	F	6/3/2016 9:30:00 AM
S523777	23	BONFRM	34	SMS Received	Attended	F	5/4/2016 9:30:00 AM
S526865	34	BONFRM	38	SMS Received	Attended	F	4/29/2016 9:30:00 AM
S527901	37	BONFRM	38	SMS Received	Attended	F	4/29/2016 9:30:00 AM
S540459	31	BONFRM	27	SMS Received	Attended	M	5/2/2016 9:30:00 AM
S540515	36	BONFRM	27	SMS Received	Attended	M	5/2/2016 9:30:00 AM
S540598	3	BONFRM	27	SMS Received	Attended	F	5/2/2016 9:30:00 AM
S540616	24	BONFRM	27	SMS Received	Attended	M	5/2/2016 9:30:00 AM
S540648	65	BONFRM	28	SMS Received	Attended	F	5/3/2016 9:30:00 AM
S541020	8	BONFRM	27	No SMS	No-Show	M	5/2/2016 9:30:00 AM
S541427	10	BONFRM	28	SMS Received	Attended	F	5/3/2016 9:30:00 AM

Page 1 Page 2 Page 3 Page 4 Page 5 Page 6 Page 7 Appointment Details Advanced Analysis +

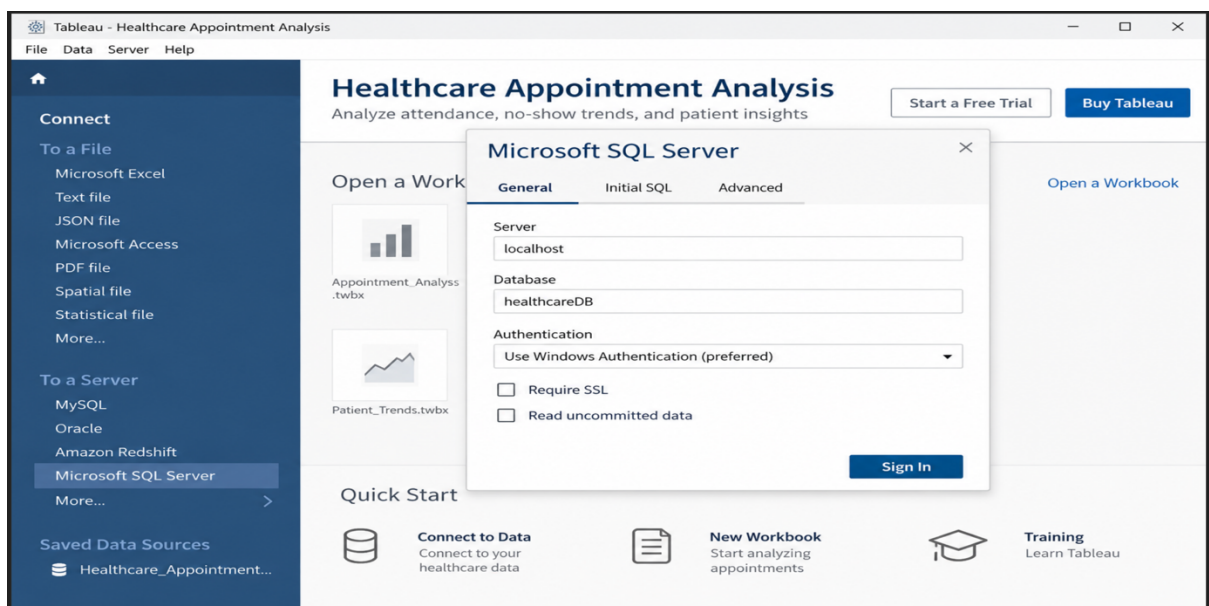
8 of 9

9. Advanced No-Show Rate Analysis

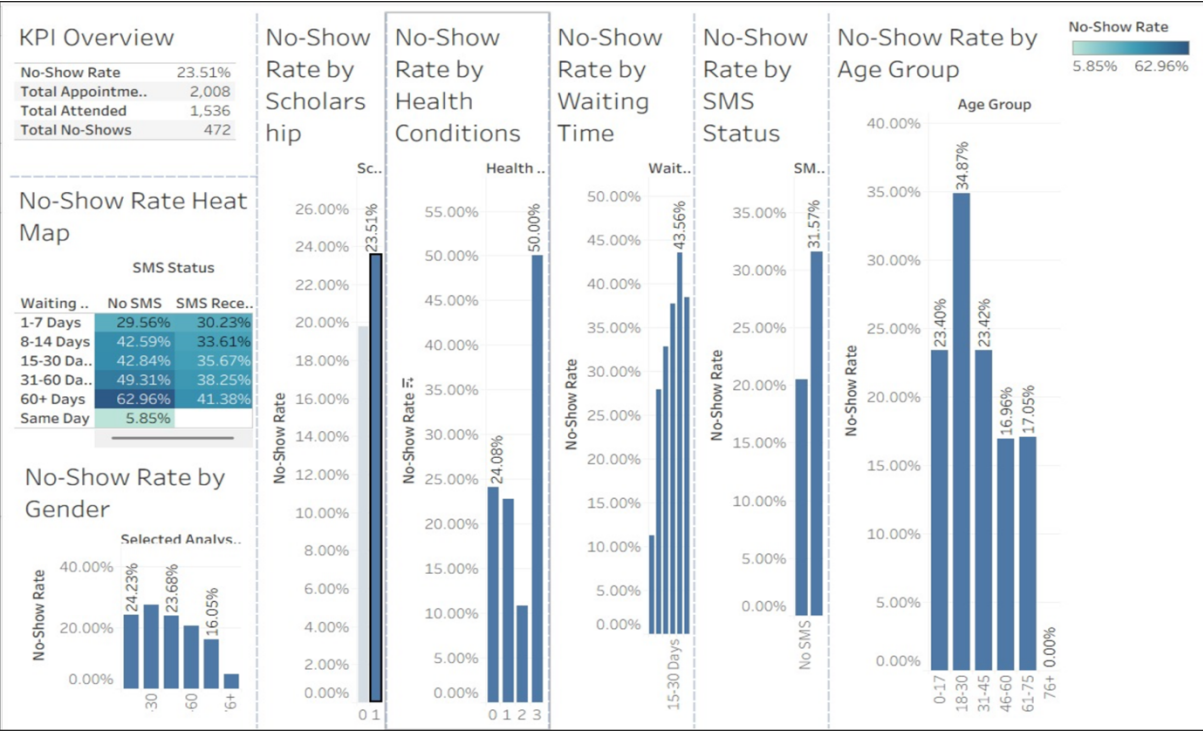


TABLEAU

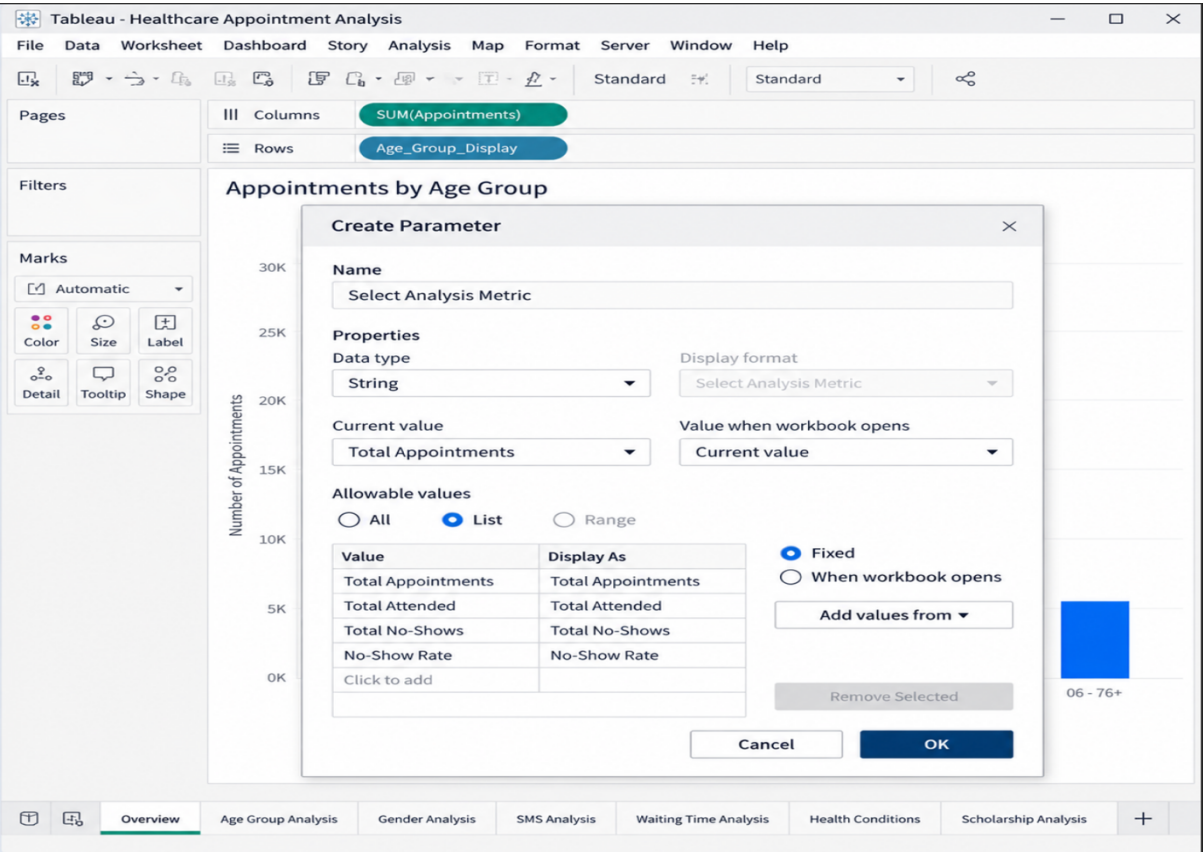
Import the data from sql server



FINAL DASHBOARD:



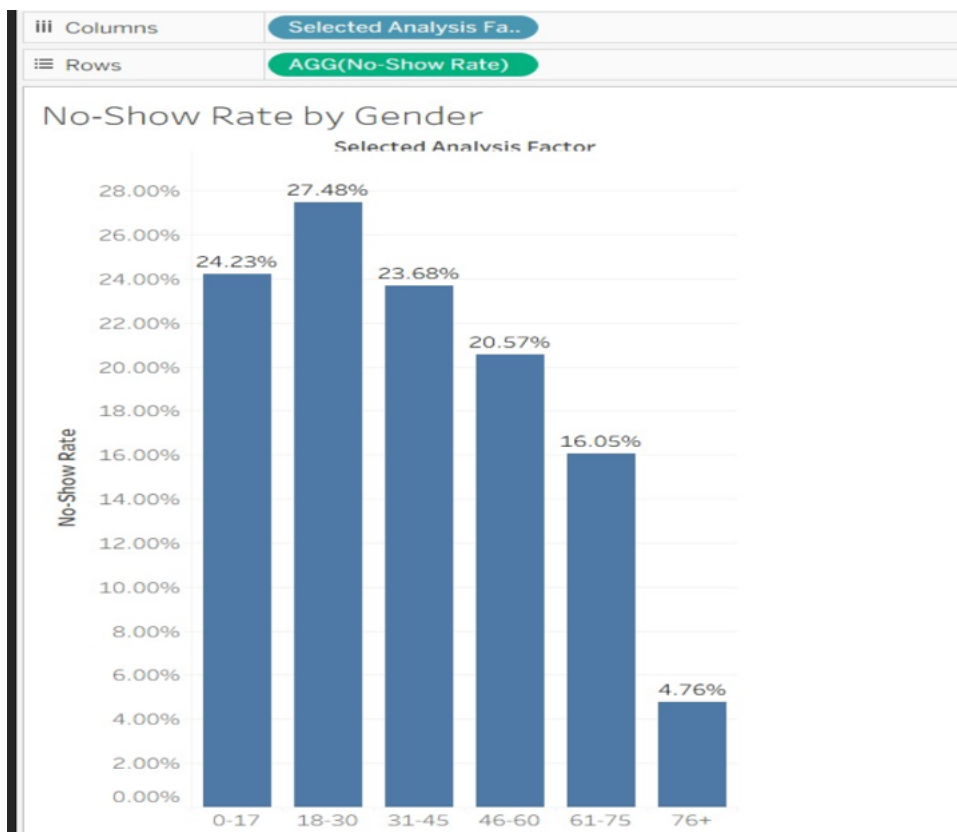
Build Parameters:



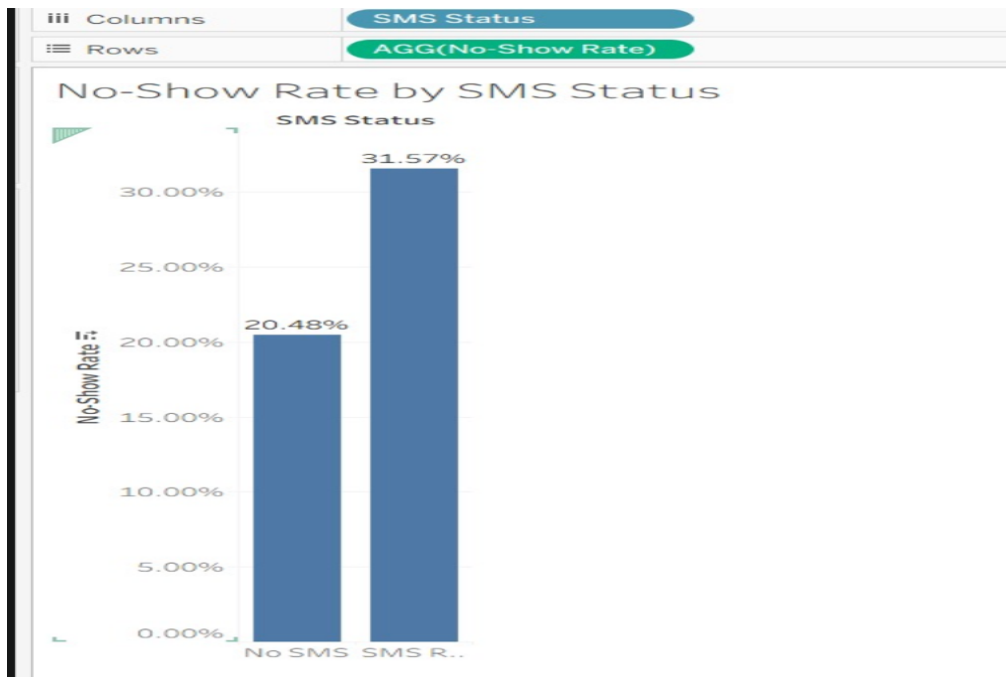
1.No-Show Rate by Age Group



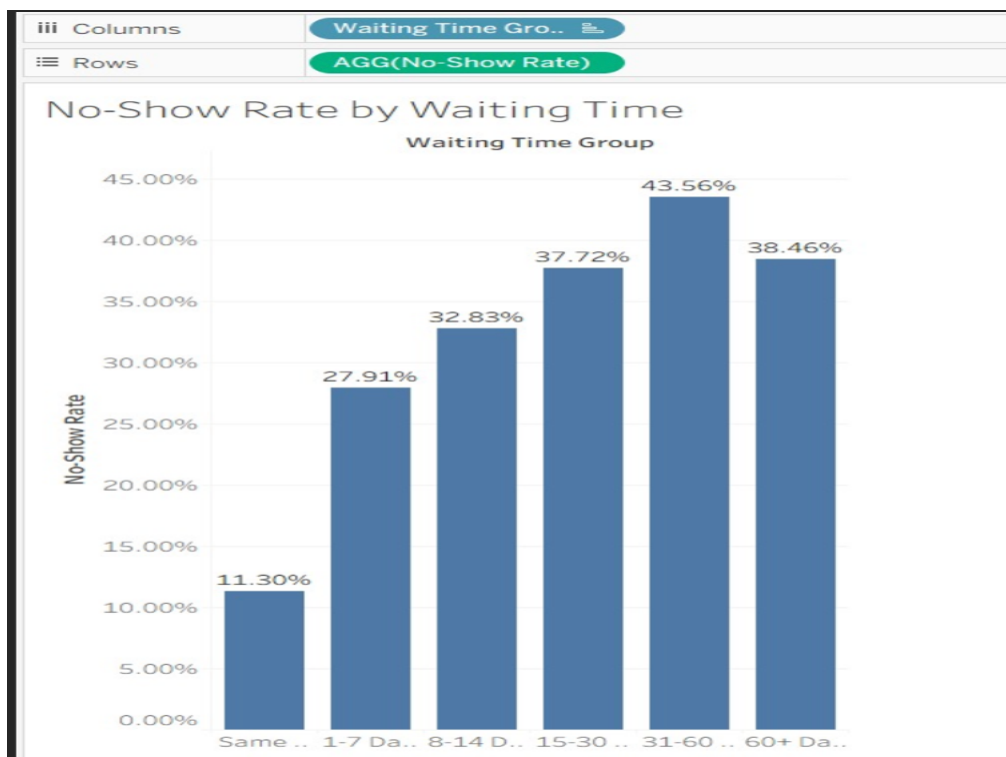
2.No-Show Rate by Selected Analysis Factor



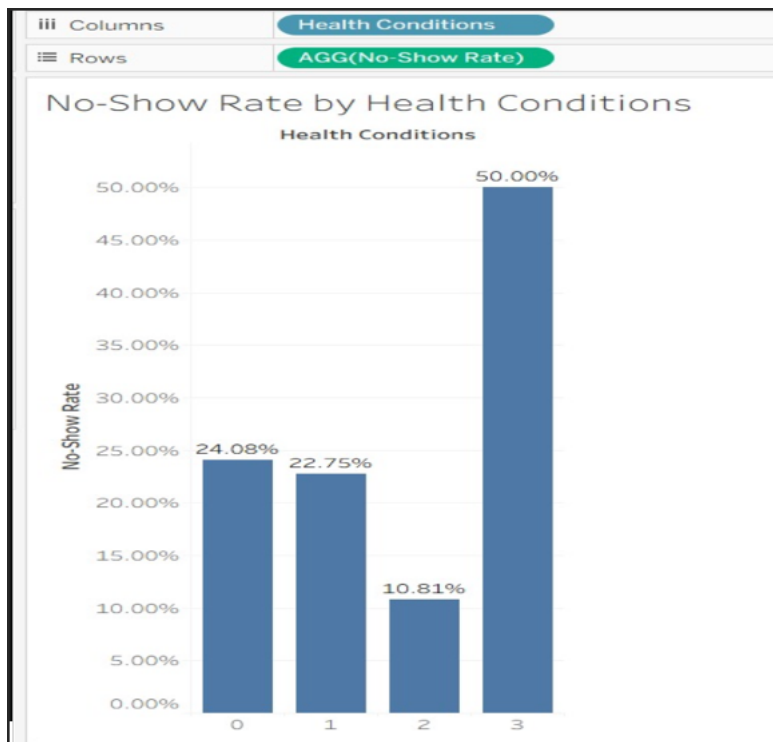
3.No-Show Rate by SMS Status



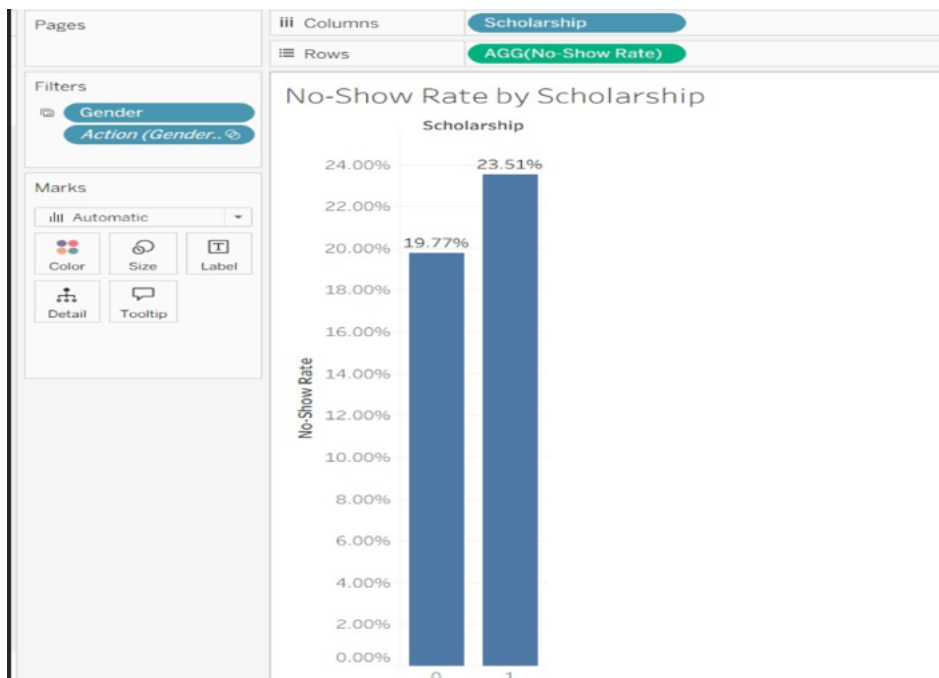
4.No-Show Rate by Waiting Time



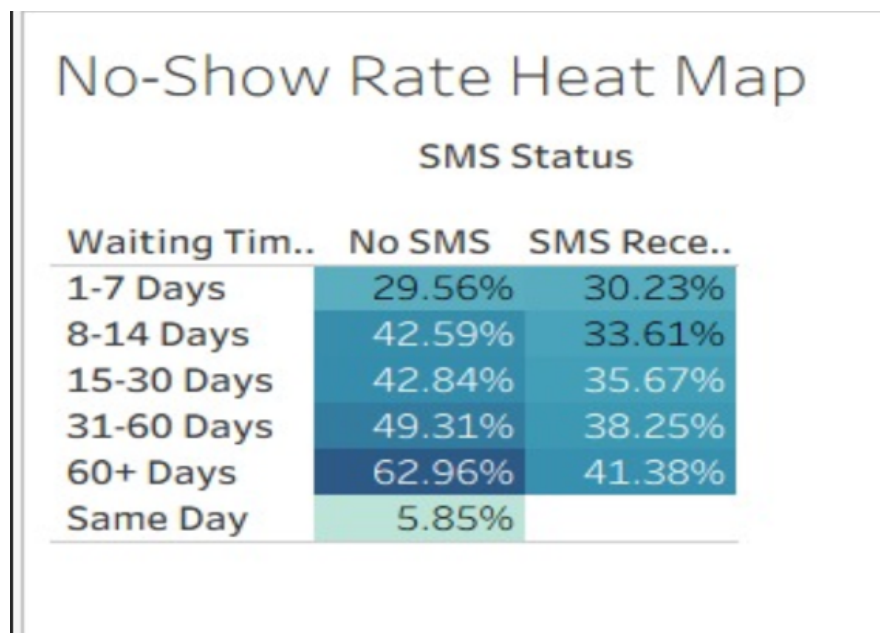
5.No-Show Rate by Health Conditions



6.No-Show Rate by Scholarship



7.No-Show Rate Heat Map



Observed Insights

- The dashboard contains 110,521 medical appointments, with 88,207 attended appointments and 22,314 no-shows.**
The overall no-show rate is approximately **20.19%**, while the attendance rate is about **79.81%**.
This indicates that around one-fifth of the scheduled appointments were missed.
- No-show rates vary noticeably across age groups, with younger and middle-aged groups showing higher no-show levels.**
The **18–30 age group** records the highest no-show rate at approximately **24.61%**, followed by **31–45** at **21.37%** and **0–17** at **21.90%**.
The **61–75 group** shows a comparatively lower rate of about **14.86%**, indicating differences in appointment attendance across age categories.
- Waiting time shows a strong relationship with appointment no-shows.**
Same-day appointments have the lowest no-show rate at approximately **11.30%**, while the rate increases to **27.91% for 1–7 days**, **32.83% for 8–14 days**, **37.72% for 15–30 days**, and reaches about **43.56% for 31–60 days**.
This indicates that appointments scheduled farther in advance tend to have higher observed no-show rates.
- SMS status and scholarship status also show differences in observed no-show rates.**
Appointments marked **SMS Received** show a no-show rate of approximately **31.57%**, compared with **20.48% for No SMS**. Scholarship appointments show approximately **23.51%**, compared with **19.77% for non-scholarship appointments**.
These differences indicate that attendance patterns vary across support and reminder categories, although the dashboard shows association rather than causation.

